SAMPLING DECISIONS

NILAM SANGROULA

SAMPLING TERMINOLOGY

- Sample
- Population or Universe
- Census
- Population element
- Sampling Frame

SAMPLE

- A sample, in the context of scientific research and statistics, is a representative subset of a population.
  - A survey involving the sleep habits of university students, would be hard to collect data from all current students
  - A researcher is researching on the effects of sound pollution in Asia.
- To get around that problem, researchers access a sample group.
- Characteristics of the sample should match those of the population so that the outcome of an experiment or survey conducted on a sample would be replicable if it were possible to research the whole population.

POPULATION

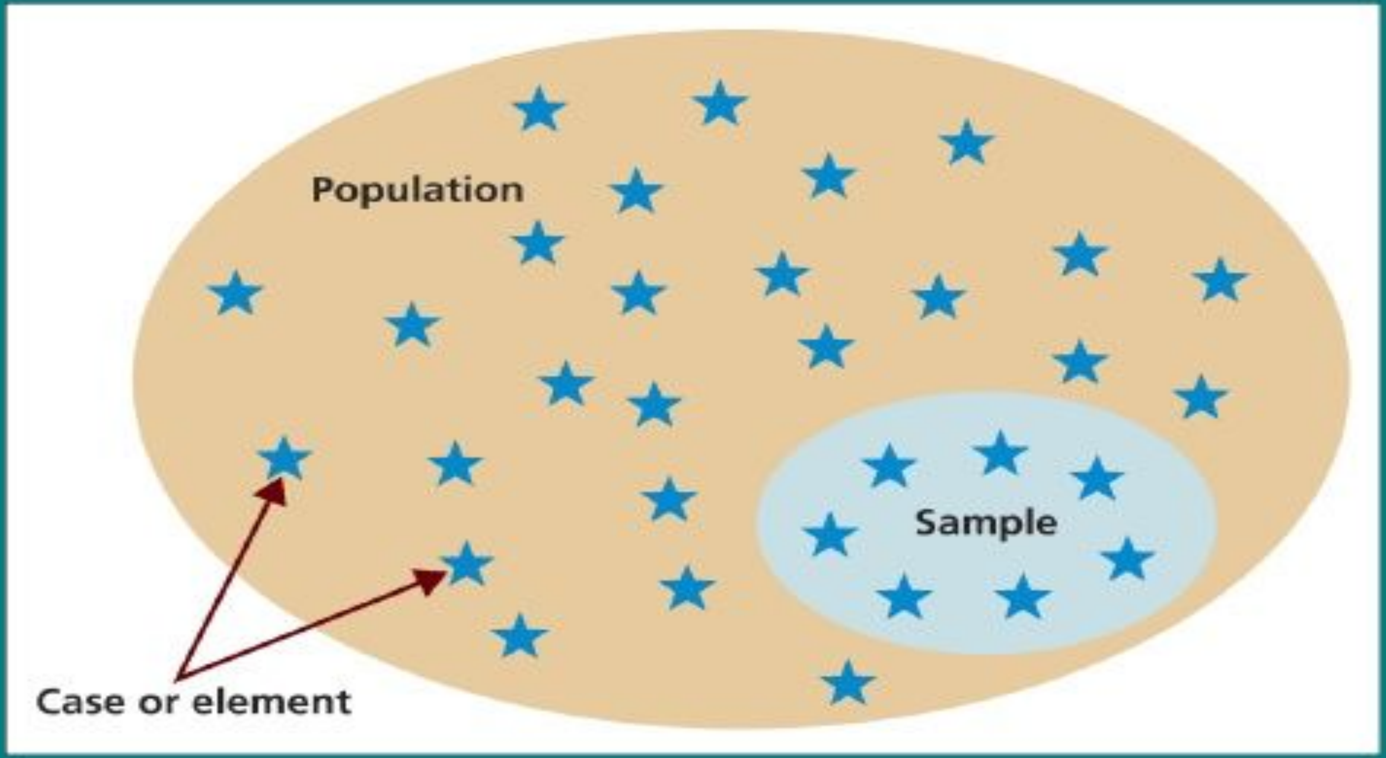
- A population is the total collection of elements about which we wish to make some inferences.
- Any complete group
  - People
  - Hospitals

CENSUS

- Investigation of all individual elements that make up a population. A census is a count of all the elements in a population.
- Population census 2021
  - Total number of households in the country is 6,666,937
  - There are 129 languages spoken as mother tongue reported in census 2021.
  - About 2.2 percent of the total population is found with some kind of disability.
  - Overall literacy rate (for population aged 5 years and above) has

Selecting samples

Population, sample and individual cases



Source: Saunders *et al.* (2009)

SAMPLING FRAME

- A sampling frame is a list or other device used to define a researcher's population of interest. The sampling frame defines a set of elements from which a researcher can select a sample of the target population.
- The difference between a population and a sampling frame is that the population is general and the frame is specific.
  - Population: Students in IELTS class.
    Sampling Frame: Ram, Hari, Sita, Gita, Laxmi, Rita, Prita, Kate, Aishwariya, Salman, Rajesh, Bob, Narayan, Paru, Dev, Swastika, munu, Vanessa, Yasmin.
  - Sample: Laxmi, Salman, Bob, Swastika, Vanessa

CONT.

- **Population**: Birds that are pink.
- **Sampling Frame**:
    - Brown-capped Rosy-Finch.
    - White-winged Crossbill.
    - American Flamingo.
    - Roseate Spoonbill.
    - Black Rosy-Finch.
    - Cassin’s Finch.
- Sample:
    - Brown-capped Rosy-Finch.

SAMPLING

- The process of using a small number of items or parts of larger population to make a conclusions about the whole population. Hence,
- Sampling is the act, process, or technique of selecting a suitable sample, or a representative part of a population for the purpose of determining parameters or characteristics of the whole population.

3 FACTORS THAT INFLUENCE SAMPLE REPRESENTATIVENESS

- Sampling procedure
- Sample size
- Participation (response)

WHEN YOU SAMPLE THE ENTIRE POPULATION?

- When your population is very small
- When you have extensive resources
- When you don’t expect a very high response

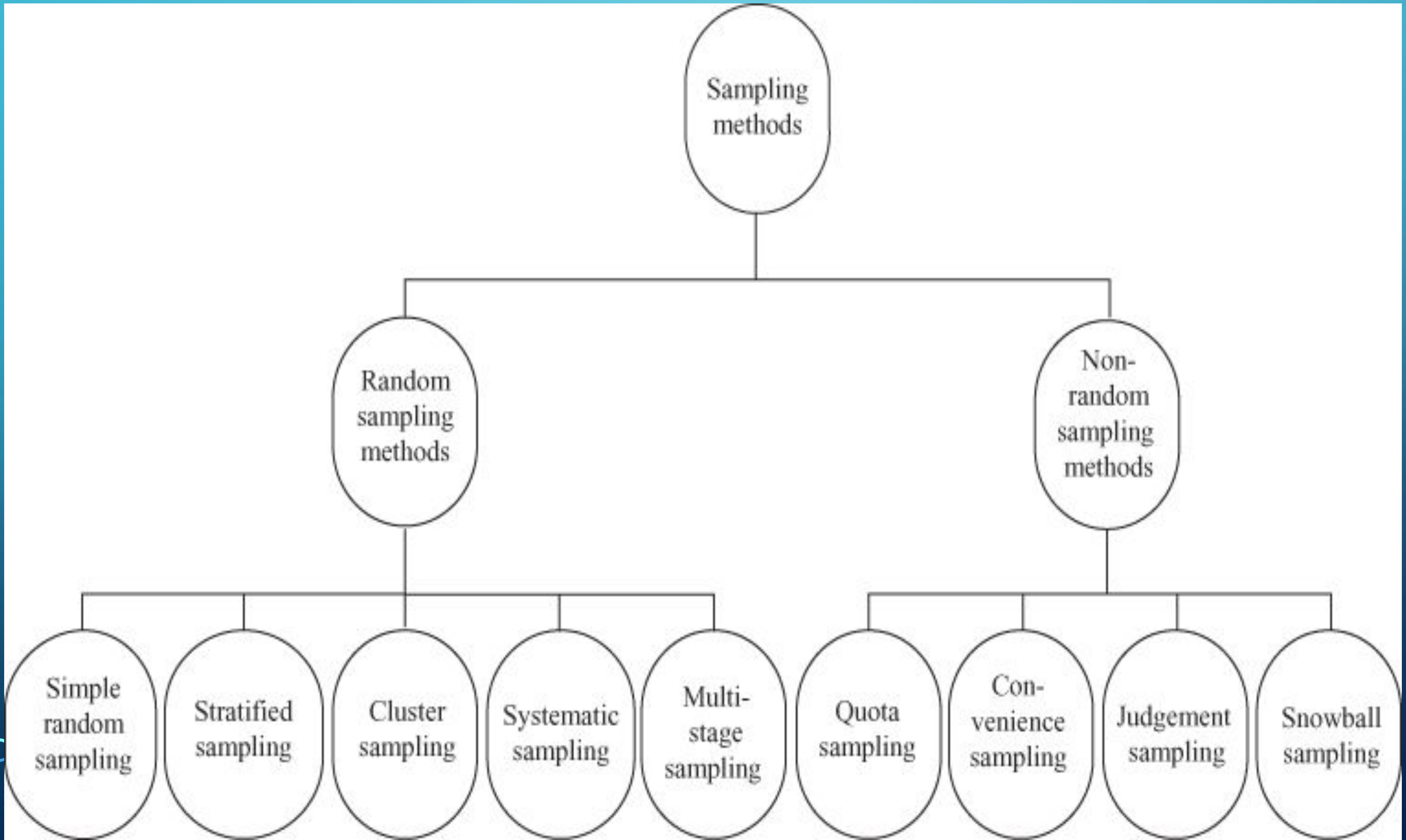
WHY IS SAMPLING ESSENTIAL?

- Sampling saves time.
- Sampling saves money.
- When the research process is destructive in nature, sampling minimizes the destruction.
- Sampling broadens the scope of the study in light of the scarcity of resources.
- It has been noticed that sampling provides more accurate results, as compared to census because in sampling, non-sampling errors can be controlled more easily. (non-response error, measurement error, interviewer error, adjustment error, processing error, questions' ambiguity)
- In most cases complete census is not possible and, hence, sampling is the

SAMPLING DESIGN PROCESS

Defining the target population
Specifying the Sampling frame
Specifying the sampling unit
Selection of the sampling method
Determination of sample size
Specifying the sampling plan
Selecting the sample

APPLICATION OF SAMPLING



RANDOM SAMPLING

- In random sampling each unit of the population has the same probability (chance) of being selected as part of the sample.
- In random sampling, the chance factor comes into play in the process of sample selection.
- Random Sampling are:
    - Simple random sampling
    - Stratified Random Sampling
    - Cluster (or Area) Sampling
    - Systematic (or Quasi-Random) Sampling

SIMPLE RANDOM SAMPLING

- In simple random sampling, each member of the population has an equal chance of being included in the sample.
- Simple random sampling is the most common method of selecting a sample from the population.
- In the simple random sampling method, first, a complete list of all the members of the population is prepared.
- Each element is identified by a distinct number (say from 1 to N).
- Then n items are selected from a population of size N using the random number generator.
- Random number generator is usually a computer program that generates

Stratified Random Sampling

- Stratified random sampling is based on the concept of homogeneity and heterogeneity. In stratified random sampling, elements in the population are divided into homogeneous groups called strata. Then, researchers use the simple random sampling method to select a sample from each of the strata.
- Stratified random sampling is also called proportional random sampling or quota random sampling.
- For example, a company that produces perfume wants to know the consumer preference for its newly launched product. For this purpose, researchers have to select a sample of 1000 consumers from a particular town with a population of 1,000,000. This population contains people from different age groups, education, regions, religion, etc. These groups may have different reasons for preferring a brand. Taking 1000 people randomly from the population will not lead to an accurate result because they may not be true representatives of the population. So, instead of selecting people directly from the population, we need to divide this heterogeneous population into homogenous groups, and then simple random sampling procedure can be used to obtain the samples from those homogenous groups.

Cluster (or Area) Sampling

- In cluster sampling, we divide the population into non-overlapping areas or clusters. It might seem as there is no difference between stratified sampling and cluster sampling. This is not true; in fact there is a well-defined difference between stratified sampling and cluster sampling.
    - For example, a fast-moving-consumer-goods company wants to launch a new product and wants to conduct a market study. For this, the country can be divided into clusters of cities and then individual consumers within cities can be selected for the survey.

Systematic (or Quasi-Random) Sampling

- In systematic sampling, sample elements are selected from the population at uniform intervals in terms of time, order, or space. For obtaining samples in systematic sampling, first of all, a sampling fraction is calculated.
  - For example, a researcher wants to take a sample of size 30 from a population of size 900 and s/he has decided to use systematic sampling for this purpose. As the first step, s/he has to calculate a sample fraction k, which is N/n, where N is the total number of units in the population and n is the sample size.
  - So, in this case, sample fraction will be 900/30=30, For obtaining the sample, the first member can be selected randomly that falls below 30 and after that every 30th member of the population is included in the

Multi-Stage Sampling

- As the name indicates, multi-stage sampling involves the selection of units in more than one stage.
- The population consists of primary stage units and each of these primary stage units consists of secondary stage units.
- In the process of multi-stage sampling, first, a sample is taken from the primary stage units and then a sample is taken from the secondary stage units.
  - For example, a researcher wants to select 100 urban households from the entire country and wants to use multi-stage sampling for this. For this purpose, s/he may first select 4 provinces from the country as the primary sampling unit. During the second stage, 20 districts from these 4 provinces may be selected. Finally, 5 households from each district may be randomly selected. Thus, a researcher will obtain the required 100 urban households in two stages.

NON-RANDOM SAMPLING

- Sampling techniques where the selection of the sampling units is not based on a random selection process are called non-random sampling techniques.
- In the selection of the sample units, the probability (of being included in the sample) is not used, which is why these techniques are also termed as non-probability sampling techniques.
- The commonly used non-random sampling techniques are:
    - Quota sampling
    - Convenience sampling
    - Judgement sampling

Quota sampling

- Quota sampling in some cases is similar to stratified random sampling.
- In Quota sampling, certain subclasses, such as age, gender, income gap and educational level are used as strata.
- In quota sampling, researchers use non-random sampling method to gather data from one stratum until the required quota fixed by the researcher is fulfilled.
  - A researcher wants to select a sample of 1000 from a population of 50,000. this population contains of 10,000 males and 40,000 females.
  - So in a sample of 1,000 people, the researcher will select 200 males and 800 females as per the population proportion.

CONVENIENCE SAMPLING

- Sometimes known as grab or opportunity sampling or accidental or haphazard sampling.
- A type of non-probability sampling which involves the sample being drawn from that part of the population which is close to hand. That is, readily available and convenient.
- The researcher using such a sample cannot scientifically make generalizations about the total population from this sample because it would not be representative enough.
  - For example, if the interviewer was to conduct a survey at a shopping center early in the morning on a given day, the people that s/he could interview. Sampling can be taken at different times of day and several times per week.

CONT.

- Use results that are easy to get

JUDGEMENT SAMPLING

- In judgement sampling, the selection of the sampling unit is based on the judgement of a researcher.
- In some cases, researchers believe that they will be able to select a more representative sample by using their judgement, which will be time and cost efficient and more accurate than simple random sampling.
- There is no well-defined scientific method which can tell us that how one person's judgement is better than another person's judgement. Generally, judgement sampling is useful when a sample size is small and demands a specific answer.

SNOWBALL SAMPLING

- In snowball sampling, survey respondents are selected on the basis of referrals from other survey respondents.
- A snowball collects ice particles when it rolls on ice. Similarly, in snowball sampling, a researcher uses a respondent to collect information about another respondent.
- When information about the subjects is not directly available, a researcher identifies a person who will be able to provide details of other respondents whose profile will fit the study.
- Through referrals, respondents can be located easily, which could otherwise be a difficult and expensive exercise.

SAMPLING ERRORS

- Sampling error has the origin in sampling itself.
- When the sample happens to be a true representative of the population, there is no problem. Sampling errors occur when the sample is not a true representative of the population.
- In complete enumeration, sampling errors are not present because in complete enumeration sampling is not being done.
- Sampling errors can occur due to some specific reasons. Some times sampling errors occur due to faulty selection of the sample.
- Secondly, some times due to the difficulty in selection a particular sampling unit, researchers try to substitute that sampling unit with another sampling unit which is easy to be surveyed.

NON-SAMPLING ERRORS

- All errors other than sampling errors can be included in the category of non-sampling errors. The following are some common non-sampling errors:
    - Faulty Designing and Planning of Survey
    - Response Errors
    - Non-Response Bias
    - Errors in Coverage
    - Compiling Error and Publication Error

METHODS OF DETERMINING SAMPLE SIZE

- Perhaps the most frequently asked question concerning sampling is, "What size sample do I need?"
- The answer to this question is influenced by a number of factors, including the purpose of the study, population size, the risk of selecting a "bad" sample, and the allowable sampling error.

THE LEVEL OF PRECISION

- The level of precision, sometimes called sampling error, is the range in which the true value of the population is estimated to be.
- This range is often expressed in percentage points, (e.g., ±5 percent)
- If a researcher finds that 60% of farmers in the sample have adopted a recommended practice with a precision rate of ±5%, then s/he can conclude that between 55% - 65% of farmers in the population have adopted the practice.

A SIMPLIFIED FORMULA

- Yamane (1967:886) provides a simplified formula to calculate sample sizes.
- $n=N/(1+Ne^2)$
- E=level of precision

- Suppose our evaluation of farmers' adoption of the new practice only affected 2,000 farmers. The sample size that would be……………

- $n = N/1+N(e)^2$
- $n = 2000/1+2000(.05)^2$
- n= 333 farmers

Cochran's Sample Size Formula

- The Cochran formula allows you to calculate an ideal sample size given a desired level of precision.
- Cochran's formula is considered especially appropriate in situations with **large populations.**
- The Cochran formula is:
- $$n_0 = \frac{Z^2pq}{e^2}$$

- Where,
- e is the desired level of precision (i.e. the margin of error)
- p is the (estimated) proportion of the population with a particular characteristic
- q is 1 – p

- Suppose you are doing a study on how many households buy kitchen chores from Satashi store Birtamode. We don’t have much information on the subject to begin with, so we’re going to assume that half of the families visit Satashi store. So $p = 0.5$. Now let’s say we want 95% confidence and at least 5% precision. A 95 % confidence level gives us Z values of 1.96, per the normal tables, so we get,
- $((1.96)^2 (0.5) (0.5)) / (0.05)^2 = 385.$

- Margin of error, E=0.05
- Confidence level, CI=0.95
- Significance level, α=1−CI=0.05
- The critical value of z:
- z_{crit} = zα/2
- = z 0.05/2
- =z 0.025
- = 1.96   Using z table

- Let p is the population proportion. P= 0.5
- The sample size:

$$n_0 = \frac{Z^2pq}{e^2}$$

- $n_0 = ((1.96)^2\ (0.5)\ (0.5))\ /\ (0.05)^2$
- $= 385.$